

Multiple Choice (10 marks)

1. Which of the following vitamins is required for fatty acid β -oxidation?
 - (a) Niacin
 - (b) Riboflavin
 - (c) Thiamin
 - (d) Vitamin C
2. Which of the following will increase in vitamin D deficiency?
 - (a) The plasma concentration of prothrombin
 - (b) The plasma concentration of preprothrombin
 - (c) The plasma activity of alkaline phosphatase
 - (d) The plasma activity of acid phosphatase
3. Which of these factors increases the risk for postmenopausal breast cancer?
 - (a) Red meat
 - (b) Dietary fat
 - (c) Fish
 - (d) Obesity
4. What is leptin resistance?
 - (a) Deficiency of leptin caused by a mutation in the leptin gene
 - (b) Lack of leptin release from the adipose tissue
 - (c) High concentrations of leptin without reduction in energy intake
 - (d) Low concentrations of leptin with increased energy intake
5. What enzyme is responsible for copying methylation marks from the parental to the daughter strand of DNA during replication?
 - (a) DNMT1
 - (b) DNMT3a
 - (c) DNMT3b
 - (d) DNMT3L

True / False (10 marks)

Answer True or False

6. True or False: Adipose tissue is the largest endogenous substrate source in the body, providing energy during prolonged periods of fasting.
7. True or False: Food balance sheets provide detailed information on food waste and consumption distribution within a country.
8. True or False: Iodine and vitamin C deficiencies are known to cause anaemia in individuals across various age groups.
9. True or False: Lactobacilli are primarily responsible for the initiation of dental caries in the mouth.
10. True or False: ATP is the immediate source of chemical energy used by skeletal muscles for contraction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of monitoring maternal weight gain during pregnancy and analyze how it impacts both maternal and fetal health.
12. Discuss the structure and function of plasma lipoproteins, focusing on their hydrophobic core and its significance in transporting lipids within the body.

End of Paper